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Interfacial configuration impact on Rashba spin splitting, excitonic radiative lifetime and band edge alignment in Janus monolayer based heterostructures

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In this work, we have explored two emerging Janus monolayers BiTeBr and SbTeBr, while constructing their corresponding heterostructures by varying the interfacial terminations. We have determined band edge alignment and excitonic radiative lifetime for four different heterostructures, which turned out to be governed by the stacking pattern and different terminations at the interfaces. We have considered many-body perturbation theory (MBPT) in addition to the ground state electronic structure calculations based on density functional theory (DFT) formalism. We have also determined the charge carrier mobility and photo-conversion efficiency (PCE) of the four possible heterostructure configurations, which are found to be sensitive depending upon the termination at the interface, as also being reflected in the Rashba spin splitting, excitonic radiative lifetime and band edge alignment. The constructed heterostructures are found to be thermodynamically and dynamically stable, while all of them show type-II band edge alignment. The photo-conversion efficiency varies significantly, ranging from 2.39% to 17.27% due to the enhancement of effective charge carrier separation originating from interlayer bright excitons with extended lifetimes reducing radiative decay.

1 Introduction

Since the successful exfoliation of graphene,[1] two-dimensional (2D) materials have garnered significant interest for optoelectronic and energy applications. A diverse range of 2D systems including silicene, germanene, transition metal di-chalcogenides (TMDCs),[2] transition metal tri-chalcogenides (TMTCs),[3] transition metal nitride and carbide MXenes,[4,5] metal mono-chalcogenides (MMCs),[6] antimonene,[7] boron nitrides,[8] and Janus materials[9,10] have been explored both theoretically and experimentally. Their unique properties, such as tunable band gaps, high electronic conductivity, strong light–matter interaction, thermal and mechanical stability, and flexibility,[11–14] make them excellent candidates for ultrathin photovoltaic (PV) and optoelectronic devices. The weak van der Waals interaction among the layers allows the 2D monolayers to be easily exfoliated from their bulk counterparts. These layers can also be stacked upon another layer in order to construct heterostructures, which further leverages a plethora of additional composite materials for a wide range of applications, which was limited in the case of only monolayers. By combining monolayers to form heterostructures, the properties of the materials can be tuned to enhance the performance of solid-state devices.[5,15,16]

Layered materials having two different facets terminated with two different elements, known as Janus structures, are emerging as next-generation 2D materials. These materials consist of a central atomic layer with metal atoms, and the outer layers consist of different atoms along the *c*-axis. Due to the presence of the atoms in the top and bottom layers with different electronegativities and charge separation, a large built-in electric field in each monolayer[10] is created. The absence of symmetry both in-plane and out-of-plane in Janus monolayers adds another degree of freedom to gain control over their characteristics.[17] TMDC-based Janus materials, initially predicted theoretically,[18] are formed by substituting one chalcogen layer with a different chalcogen and have since been successfully synthesized.[19] Group V-a of the Janus monolayer is composed of halides and chalcogens on the outer layer and bismuth or antimony atoms in the center. These materials exhibit excellent piezoelectric characteristics, while the bismuth-based materials also show excellent Rashba splitting, which has been reported in previous studies.[20,21] Bismuth-based Janus materials are successfully synthesized by various methods, such as chemical vapor deposition and also mechanical exfoliation from bulk crystals.[21,22] Considering these unique characteristics and the theoretically

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suggested stability, there is room for the synthesis of antimony-based Janus materials.[23,24] The difference in the chemical environment on different sites of the monolayers provides an opportunity to build different configurations of heterostructures, giving rise to different properties. This has been advantageous for tuning the properties of heterostructures as compared to other 2D materials.

The commercialization of large-scale solar energy requires photovoltaic (PV) technology which is inexpensive and extremely efficient. Due to their low manufacturing costs, conventional silicon (Si)-based solar cells have been used in the industry for the past few decades.[25] However, with the introduction of hybrid perovskites in 2009, these materials are being explored to be used in solar cell applications.[26,27] Similar efficiency has been achieved by the perovskite class of materials, as compared to the Si-based solar cells.[28] Despite this, there is still a desire for photovoltaic (PV) materials that are thin and flexible with substantial mechanical stability, and having the ability to withstand extreme mechanical stress under extreme conditions. The 2D materials are ideal in this context due to their high power conversion rate per weight, which can further create a scope for wearable energy-harvesting devices.[29] Heterostructures based on Janus monolayers offer enhanced photovoltaic performance. Depending on the electronic nature of the constituents, these devices are classified as excitonic (type-II semiconducting layers) or Schottky barrier (semimetal–semiconductor) solar cells.[30,31] Type-II excitonic heterostructures enable efficient interlayer exciton generation, charge transfer, and suppressed recombination, leading to improved solar energy conversion.[32–36]

In this work, we have investigated systematic electronic structure calculations by coupling density functional theory (DFT) formalism with many-body perturbation theory (MBPT) to explore the quasi-particle and excitonic characteristics of the 2D Janus monolayers BiTeBr and SbTeBr, belonging to the $P3m1$ space group, and their corresponding heterostructures. Various methods are used to study the structural, mechanical and dynamical stability of these monolayers, which are further used for the construction of four different heterostructure combinations. All of these heterostructures have been investigated further for phonon dispersion based dynamical stability similar to the monolayers. The efficiency of the photovoltaic devices is explored by understanding the band edge alignment, which is obtained from the hybrid exchange–correlation functional, while the Bethe–Salpeter equation (BSE) in addition to G_0W_0 has been incorporated to understand the many-body effects and excitonic properties, as explained in the subsequent computational methodology section.

2 Computational methodology

All the ground state calculations are carried out using Vienna *ab initio* simulation package (VASP)[37,38] based on density functional theory (DFT)[39,40] within the formalism of the projector augmented-wave method (PAW)[41,42] for describing the electron–ion interaction. The general gradient approximation (GGA) form of the Perdew, Burke, and Ernzerhof (PBE) potential is used as the exchange–correlation functional,[43] while the relativistic spin–orbit coupling (SOC)[44] is taken into consideration throughout the calculations due to the presence of heavy elements in the structures. In order to obtain complete lattice relaxation, the convergence threshold criterion of the Hellman–Feynman forces of continuous atoms is considered to be less than 2×10^{-3} eV Å^{-1}, and for self-consistency energy convergence, the convergence criteria have been set to 10^{-6} eV. The heterostructures are constructed by stacking the two different monolayers with different stacking orders and varying the interlayer distances, which are optimized to obtain the minimum energy configuration of the heterostructures. Four distinct heterostructures are taken into consideration, based on the different combinations of interfacial atoms during the stacking and the outermost atoms that create the surface. The 2D Brillouin zone has been sampled using the Gamma-centered scheme, having a k-point mesh of $11 \times 11 \times 1$, and the wave function expansion of the kinetic energy cutoff is set to 500 eV. We have considered a sufficiently large vacuum more than 15 Å along the c-axis, which would be sufficient to neglect the interactions between periodic images. To consider the interlayer van der Waals (vdW) interaction, Grimme's DFT-D3 [45] correction method has been employed throughout the calculations. We have used the Heyd–Scuseria–Ernzerhof type of hybrid exchange–correlation functional (HSE06)[46] to calculate a more accurate band structure with reliable band gap and band offset values. Electrostatic potential is calculated in the presence of dipole correction. The elastic constants for mechanical stability are obtained from the variation of energy with respect to strain, corresponding to uniaxial and biaxial deformations, and taking the second derivative of these energies. The relationship is calculated by varying the magnitude of the strain in increments of 0.5%, from −2% to 2%. The dynamical stability is determined by calculating phonon band structure using the finite displacement method using a $5 \times 5 \times 1$ supercell and $5 \times 5 \times 1$ k – point mess, as implemented in the Phonopy code.[47]

The excitonic states and the optical absorption spectra are determined using *ab initio* many-body perturbation theory with the Bethe–Salpeter equation (BSE).[48–50] This formalism expresses the excitations as electron–hole pairs:

$$(\varepsilon_{ck} - \varepsilon_{vk})A^{s}_{vck} + \sum_{v'c'k'} \left\langle vck|K_{eh}|v'c'k' \right\rangle A^{s}_{v'c'k'} = E^{s}_{x}A^{s}_{vck} \qquad (1)$$

where ε_{vck} represents the quasi-particle energies; v and c represent the valence and conduction band states. The expansion coefficients of excitonic states are A_{vck}, while their energies are E^{S}_{x}. The electron–hole interaction kernel, K_{eh}, consists of the screened direct Coulomb interaction (which is attractive) and the unscreened exchange interaction (which is repulsive). The energies and wave functions are obtained from density functional theory (DFT) calculations performed using the Quantum ESPRESSO[51] using Perdew–Burke–Ernzerhof (PBE),[52] and norm-conserving Vanderbilt pseudo-potentials[53] are utilized with 80 Ry cutoff. The quasi-particle eigenvalues are corrected from the PBE energies using G_0W_0 [54,55] approximation as implemented in the Yambo code.[56,57] The G_0W_0 quasi-particle energies and the BSE calculations are calculated on a Γ centered k-grid at $30 \times 30 \times 1$. A total of 140 and 100 bands are used for dielectric screening and self-energy, respectively. To avoid a long-range

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interaction between periodic copies along the vertical direction, we have used the coulomb cutoff screening potential in both G_0W_0 and BSE calculations.

3 Results and discussion

3.1 Structural and dynamical stability of Janus monolayers and heterostructures

The 2D Janus XTeBr (X = Bi, Sb) monolayer consists of three atomic layers stacked in the Te–X–Br sequence. The monolayer structure consists of a hexagonal phase with the space group $P3m1$, which is shown along the bird's eye view and side view for the Janus materials XTeBr in Fig. 1. After the full structural relaxation of the lattice parameters, the optimized structural parameters like lattice constants, bond lengths, bond angles, and monolayer thickness are depicted in Table 1. The lattice constant of the BiTeBr monolayer is 4.33 Å, which is slightly larger than that of the SbTeBr monolayer (4.22 Å), leading to a small lattice mismatch of 2.54%. Since this lattice mismatch is insignificant, we have constructed heterostructures from these monolayers, which will be discussed in the subsequent section. Similar to the lattice parameter, BiTeBr has a larger monolayer thickness and bond lengths between neighboring atoms due to the larger atomic radius of Bi compared to Sb.

To determine the stability of the monolayers, we have determined the formation energies, phonon dispersions, and second-order elastic constants (SOECs) for the confirmation of thermodynamical, dynamical, and mechanical stability. Initially, we determined the thermodynamic stability of the monolayers by noting the formation energy of the monolayers calculated by using the formula:

$$E_{\text{form}} = E_{\text{mono}} - (E_{\text{X}} + E_{\text{Te}} + E_{\text{Br}}) \tag{2}$$

where E_{mono} is the total energy of the monolayer. E_{X}, E_{Te} and E_{Br} are the chemical potentials of the X = Bi or Sb atom (for BiTeBr or SbTeBr), Te atom, and Br atom, respectively. The formation energy of BiTeBr and SbTeBr is found to be −1.66 eV and −1.39 eV, respectively. The negative values of E_{form} demonstrate that the formation of the structures is energetically favorable and signifies thermodynamic stability. The dynamical stability of these monolayers has also been studied by calculating the phonon band structure, which is shown in Fig. 2. The phonon dispersion displays the longitudinal acoustic (LA), transverse acoustic (TA), and out-of-plane acoustic (ZA) branches. The band structure exhibits no significant imaginary frequencies across the Brillouin zone, confirming the dynamical stability of the system. A minor negative frequency (<0.05 THz) observed in the ZA in the SbTeBr branch near the Γ point is attributed to numerical limitations such as finite supercell size and k-point sampling.[58] The phonon band dispersions show the absence of negative frequency, indicating dynamic stability. In addition to the thermodynamic and dynamical stability, the mechanical stability of the monolayers is determined by computing the elastic constants of the monolayers. The SOECs provide information about the mechanical properties of the monolayers, indicating their stability and stiffness. In the case of a 2D hexagonal structure, there are two independent SOECs, C_{11} and C_{12}. These SOECs are calculated by fitting the second derivative curve of the energy *vs.* strain plot. The equations for determining the SOECs are given by:

$$C_{11} = \frac{1}{A_0}\frac{\partial^2 E_{\text{u}}}{\partial s^2}\bigg|_{s=0}, \quad 2(C_{11} + C_{12}) = \frac{1}{A_0}\frac{\partial^2 E_{\text{b}}}{\partial s^2}\bigg|_{s=0} \tag{3}$$

In the case of both monolayers, the SOECs satisfy the Born–Huang mechanical stability condition[59,60] given in eqn (1) and reported in Table 1 for 2D hexagonal monolayers.

$$C_{11} > 0, \text{ and } C_{11} > |C_{12}| \tag{4}$$

On the basis of the atoms that form the outermost surface and atoms that participate in the vdW interaction (interface

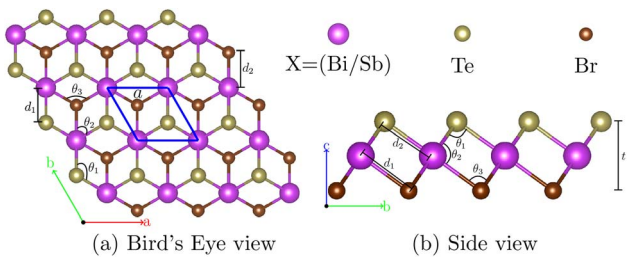

Fig. 1 Crystal structure of the 2D Janus monolayer XTeBr from (a) bird's eye view and (b) side perspectives, where X = Bi/Sb.

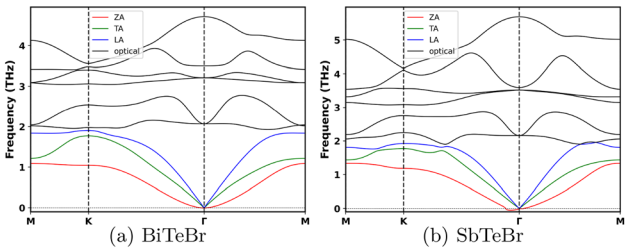

Fig. 2 Phonon band dispersions of 2D Janus monolayers (a) BiTeBr and (b) SbTeBr for determining dynamical stability. Red, green and blue bands represent ZA, TA, and LA bands respectively while black represents all optical bands.

Table 1 Geometric and elastic properties of the monolayers, including lattice constant a, thickness t, atomic distances d_1 between X and Te, d_2 between X and Br, angles (θ_1, θ_2, θ_3) as shown in Fig. 1, formation energy (E_{form}), and elastic constants (C_{11}, C_{12})

Monolayer	a (Å)	t (Å)	d_1 (Å)	d_2 (Å)	θ_1 (°)	θ_2 (°)	θ_3 (°)	E_{form} (eV)	C_{11} (N m^{-1})	C_{12} (N m^{-1})
BiTeBr	4.33	3.59	3.06	3.09	89.97	90.63	88.77	−1.66	28.62	7.2
SbTeBr	4.22	3.52	2.99	3.02	89.88	90.87	88.45	−1.39	32.14	7.85

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atom) with another monolayer, four possible orientations are taken into consideration to construct the heterostructure. To achieve a heterostructure with the minimum energy which is also dynamically stable, various stackings (shown in Fig. S2 of SI) are taken into consideration. Following complete lattice relaxation, the optimized geometries of the heterostructures with minimum energy are considered for dynamical stability. The heterostructure with positive phonon frequency, showing dynamical stability, is depicted in Fig. 3 and considered for further calculations. These heterostructures exhibit a consistent lattice parameter of approximately 4.27 Å with AB or AB′ stacking order. The binding energy is further calculated as $E_B = E_{hetero} - E_{BiTeBr} - E_{SbTeBr}$, where E_{hetero} is the total energy of the heterostructures and E_{BiTeBr}, E_{SbTeBr} are the total energy of the individual monolayers, while the negative values indicate the energetically favorable condition for the formation of the heterostructure. The heterostructure with Te–Te as the interface atom has the lowest binding energy and higher interlayer distance, and that with Br–Br as the interfacial atoms falls on the other end. The mechanical stability of the considered structures is investigated by determining the SOECs, which are reported in Table 2, and satisfying the stability condition. The charge transfer between the monolayers of the heterostructure and its projections along the z axis is shown in Fig. 4, the electron depletion is shown in cyan, and the yellow colour represents the accumulation of electrons in the 3D plot. All the four heterostructures undergo charge redistribution, influenced by the atoms involved through van der Waals interactions. In the H1 configuration, where the Te–Te atoms are involved in the vdW interaction, we see strong charge accumulation in the interface surrounded by a charge depletion layer close to the Te atoms. In the configuration H2, where Br–Br atoms are involved in the vdW interaction, we have charge depletion at the interface and also close to the Br atom. In the latter case of H3 and H4 configurations, we can see a larger charge accumulation at the interface involving the Te atom and a charge depletion layer near the interface of the Br atom. Because of the difference in charge redistribution in different configurations of the heterostructure, the electronic properties of the heterostructure vary, which is a crucial factor in photovoltaic applications.

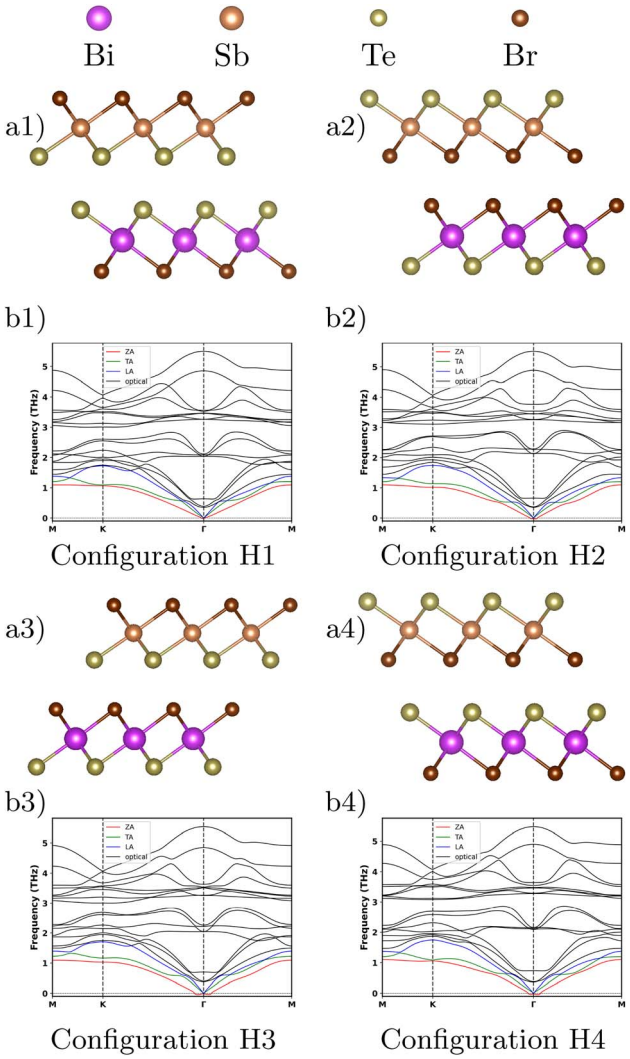

Fig. 3 Optimized geometry of different possible arrangements of the heterostructure (a1–a4) along with phonon band dispersions (b1–b4).

3.2 Electronic structures of Janus monolayers and heterostructures: presence of Rashba spin splitting

The electronic properties are investigated systematically by computing the electronic band structure as shown in Fig. 5 along with the projected density of states (PDOS) as shown in Fig. S3 of SI, and further calculating the effective mass and mobility of the charge carriers. The SOC effect is taken into consideration due to the presence of heavy elements such as Bi, Sb, and Te. Since the PBE functional underestimates the band gap, the hybrid functional HSE06 is considered to obtain an accurate band gap and offsets that are closer to the experimental value. The estimated band gaps using HSE06, PBE along with G_0W_0 are mentioned in Table 4. Since the band gap lies in the infrared region, the monolayers can absorb significant amounts of solar radiation and are suitable for photovoltaic applications. Upon visualizing the PDOS, we can observe the

Table 2 Interface atom involved in the construction of different heterostructure configurations; interlayer distance t; binding energy of the heterostructure E_B; and the SOECs C_{11} and C_{12}

	Interface atom	t (Å)	E_B (meV)	C_{11} (N m^{-1})	C_{12} (N m^{-1})
Config. H1	Te–Te	3.18	−251.3	60.38	17.06
Config. H2	Br–Br	2.99	−173.4	62.50	14.76
Config. H3	Br(BiTeBr)–Te(SbTeBr)	3.10	−230.2	63.13	17.78
Config. H4	Te(BiTeBr)–Br(SbTeBr)	3.13	−220.1	59.16	12.96

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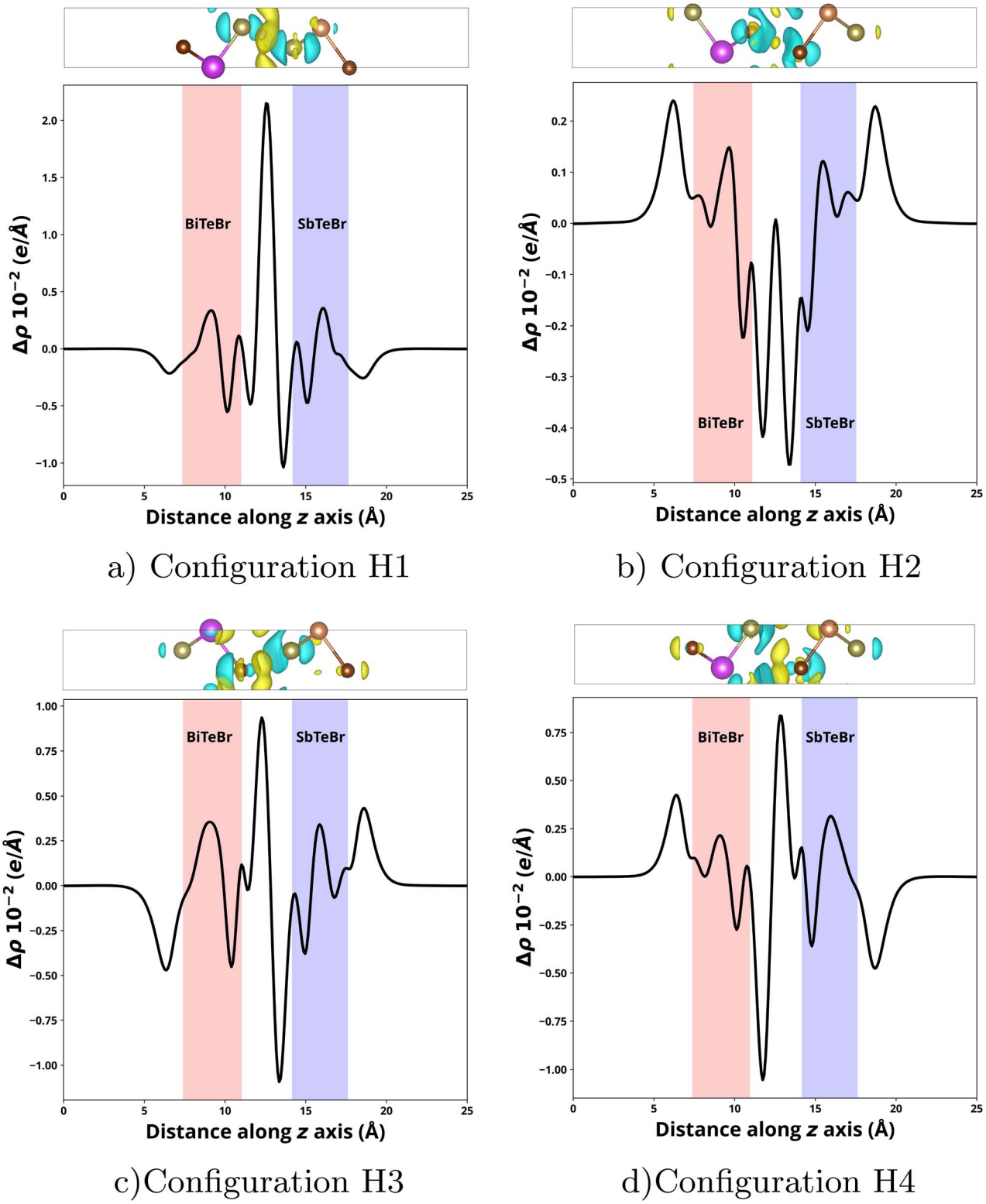

Fig. 4 The charge projection in the z-direction of several heterostructure configurations is shown with the 3D charge density difference. The charge accumulation (electron gain) and charge depletion (electron loss) are represented by the colors yellow and cyan, respectively. (a) Config. H1 constitutes both Te as the interface atoms. (b) Config. H2 constitutes both Br as the interface atoms. (c) Config. H3 constitutes the Br atom of BiTeBr and Te atom of SbTeBr as the interfacial atoms. (d) Config. H4 constitutes the Te atom of BiTeBr and Br atom of SbTeBr as the interface atoms.

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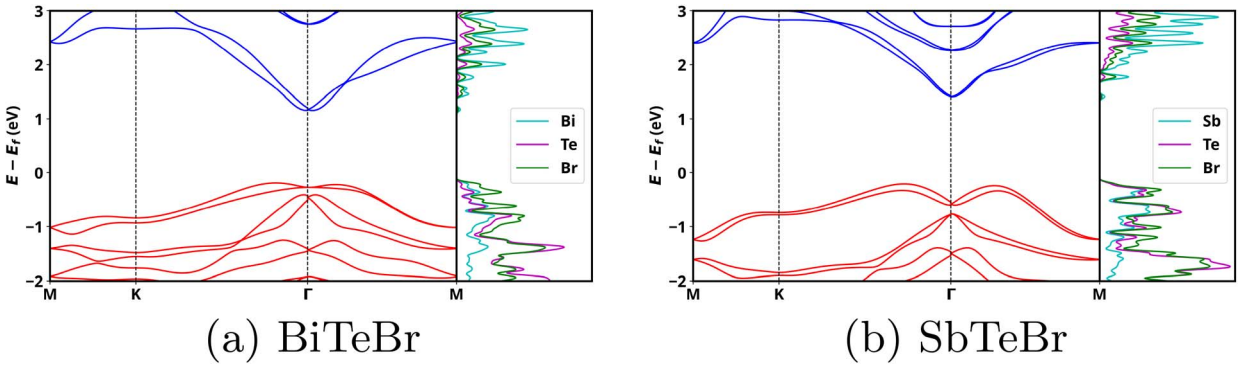

Fig. 5 Electronic band structure and atom projected density of states of (a) BiTeBr and (b) SbTeBr monolayers, determined using HSE06 type hybrid exchange–correlation functional.

major contribution from Te-5p orbitals and the minor contribution from Br-4p atoms in the valence band near the Fermi energy, while in the conduction band region, Bi-6p orbitals and Sb-5p orbitals play a major role. The VBM position in both monolayers is located near the Γ point, which is found from the band structure of the monolayers. For SbTeBr, the CBM is located at Γ, while for BiTeBr, we can observe the Rashba splitting (schematics shown in Fig. S4 of SI) in the CBM region, which slightly shifts the minima from the high symmetry Γ point. The Rashba splitting strength is defined as $(2\Delta E)/\Delta k$, which is found to be 1.256 eV Å along both Γ to K and Γ to M. The spin texture of the splitting corresponding to the bands is shown in Fig. 6. The in-plane spin component has a helical nature, and the spins in both bands are found to be in the opposite direction, which confirms the existence of Rashba-type splitting.

To obtain accurate band gap and band offsets, as demonstrated in Fig. 7, the hybrid exchange–correlation (HSE06) functional has been used to calculate the projected band structure of the layers and the atom projected density of states. Similar to monolayers, the heterostructures have their VBM and CBM located near the Γ point. The Config. H1 has a direct band gap located at the Γ point unlike monolayers or other configurations, which have a slight shift in either one or both band edge positions. Although the Bi atom of BiTeBr contributes to the CBM, this heterostructure does not exhibit the Rashba splitting behavior that is evident in the case of BiTeBr. In Config. H1 the electrostatic potential difference (shown in Fig. 8) between the vacuum region is found to be zero. The contribution to VBM is observed from the SbTeBr monolayer; even in this instance, there is no shift in the position of the VBM that was observed in the monolayer near the Γ point. The BiTeBr monolayer is seen to contribute to CBM in configurations H2 and H3, where we see the Rashba splitting. There was a slight decrease in the Rashba parameter for Config. H2, which was noted as 0.824 eVÅ and 0.976 eVÅ along the Γ–K and Γ–M directions, respectively. In Config. H3, the Rashba splitting parameter was observed to increase slightly to 1.193 eVÅ along Γ–K and along Γ–M it was observed to be 1.426 eVÅ. This variation in the Rashba parameter is analyzed by examining the electrostatic potential along the z-axis, including dipole correction, as illustrated in Fig. 8. In configuration H2, where both Te atoms terminate near the vacuum interface, the electrostatic potential difference remains close to zero. As a result, no significant internal electric field is generated, leading to a reduced Rashba parameter. In contrast, configuration H3 exhibits a larger electrostatic potential difference compared to BiTeBr, which induces a stronger internal electric field. This enhanced field contributes to an increase in the Rashba splitting parameter. SbTeBr contributes to the VBM region in both of these heterostructures. However, for Config. H2 we can also see the contribution from BiTeBr at the Γ point which is near the VBM. Unlike the first three configurations, the contribution in the CBM is by the SbTeBr monolayer, and the contribution in the VBM is by the BiTeBr monolayer. Since SbTeBr contributes predominantly to the conduction band minimum (CBM) and does not exhibit Rashba splitting on its own, the heterostructure also shows no Rashba-type splitting at the CBM. Therefore, it is worth mentioning that, in all four heterostructure configurations, the electronic band structure and band gap are governed by the terminating atoms at the interface.

The longitudinal acoustic phonon-limited carrier mobility in 2D monolayers has been calculated using deformation potential (DP) theory to understand the flow of charge carriers. This is a widely employed hypothesis used to calculate the carrier mobility in 2D materials suggested by Bardeen and Shockley.[61] The charge carrier mobility is influenced by their effective mass. To determine the effective mass and DP along the x and y directions, we have analyzed the conventional orthorhombic supercell shown in Fig. S1 of SI and the corresponding band structure shown in Fig. S5 of SI. The effective masses of electrons and holes are calculated from the electronic band structure of PBE using the expressions $1/m^*_{\mathrm{ei}} = 1/\hbar^2(\partial^2 E_{\mathrm{CBM}})/(\partial k_i{}^2)$ and $1/m^*_{\mathrm{hi}} = 1/\hbar^2(\partial^2 E_{\mathrm{VBM}})/(\partial k_i{}^2)$, respectively. In these equations, E_{CBM} and E_{VBM} represent the energies at the conduction band minimum (CBM) and the valence band maximum (VBM), while k_i corresponds to the reciprocal lattice vectors along the x and y directions. The CBM and VBM in both monolayers appear near the Γ point. Therefore, the carrier effective mass has been calculated along the Γ–X and Γ–Y directions, which corresponds to the x and y directions in real space. The deformation

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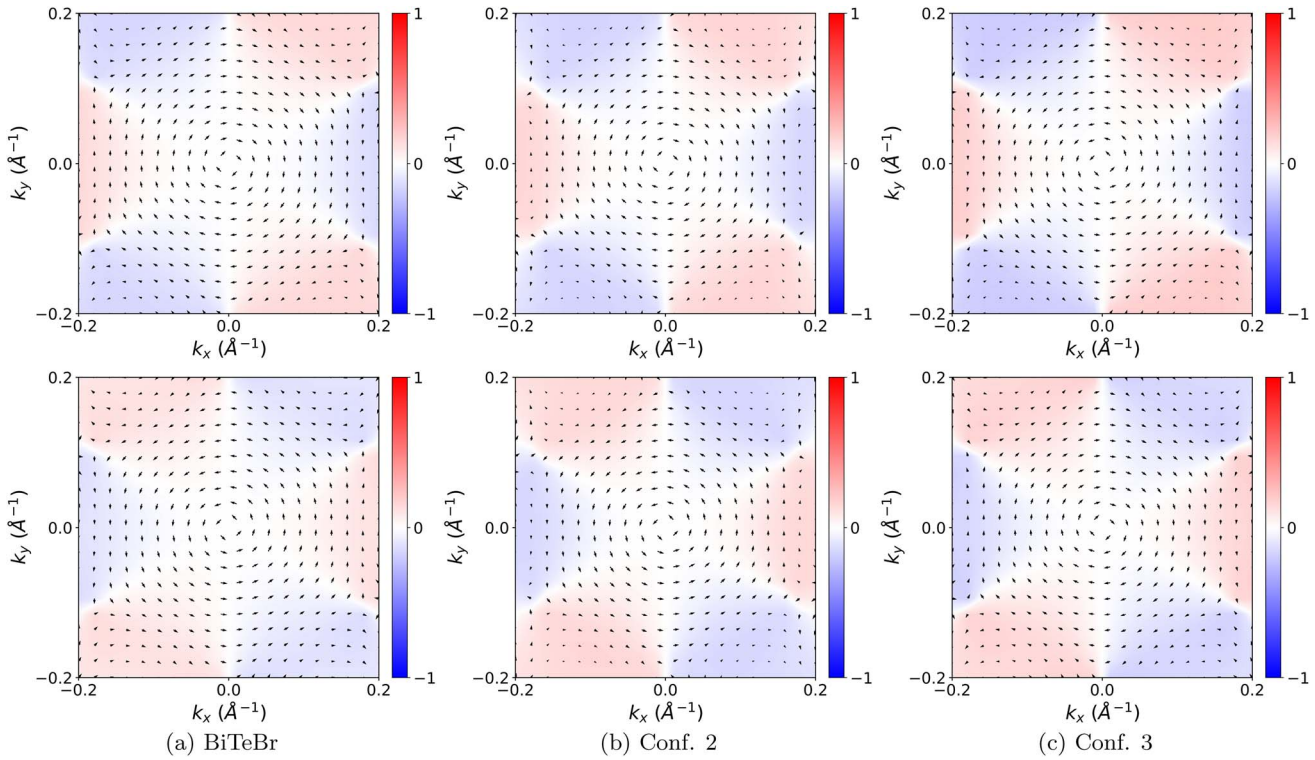

Fig. 6 Spin texture of (a) BiTeBr monolayer, (b) configuration H2, and (c) configuration H3 at CBM. The top and bottom panels correspond to the bands near the CBM at Γ point where Rashba splitting is observed. The arrows represent the in-plane components of spin texture; the out of plane components of spin texture are represented by the colour projection.

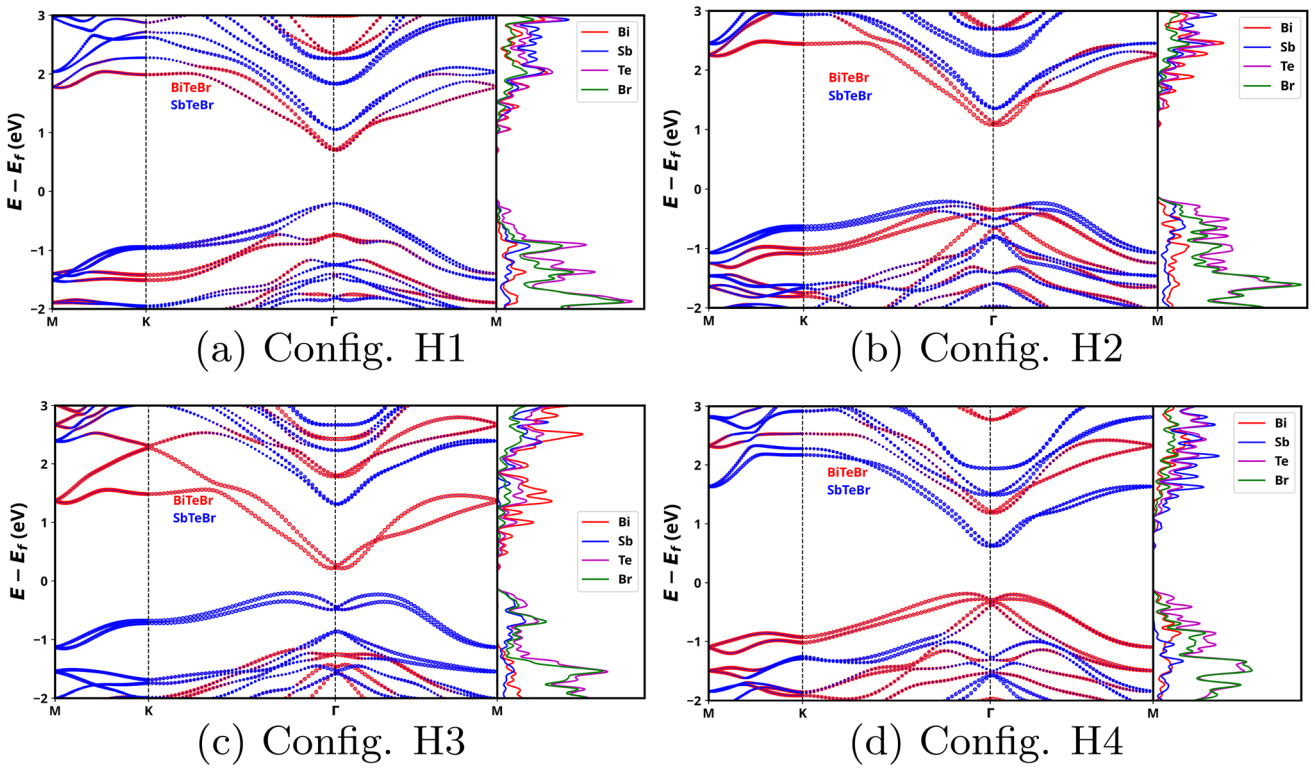

Fig. 7 Layer projected electronic band structure along with atom projected density of states of different heterostructures: (a) Config. H1, (b) Config. H2, (c) Config. H3, (d) Config. H4.

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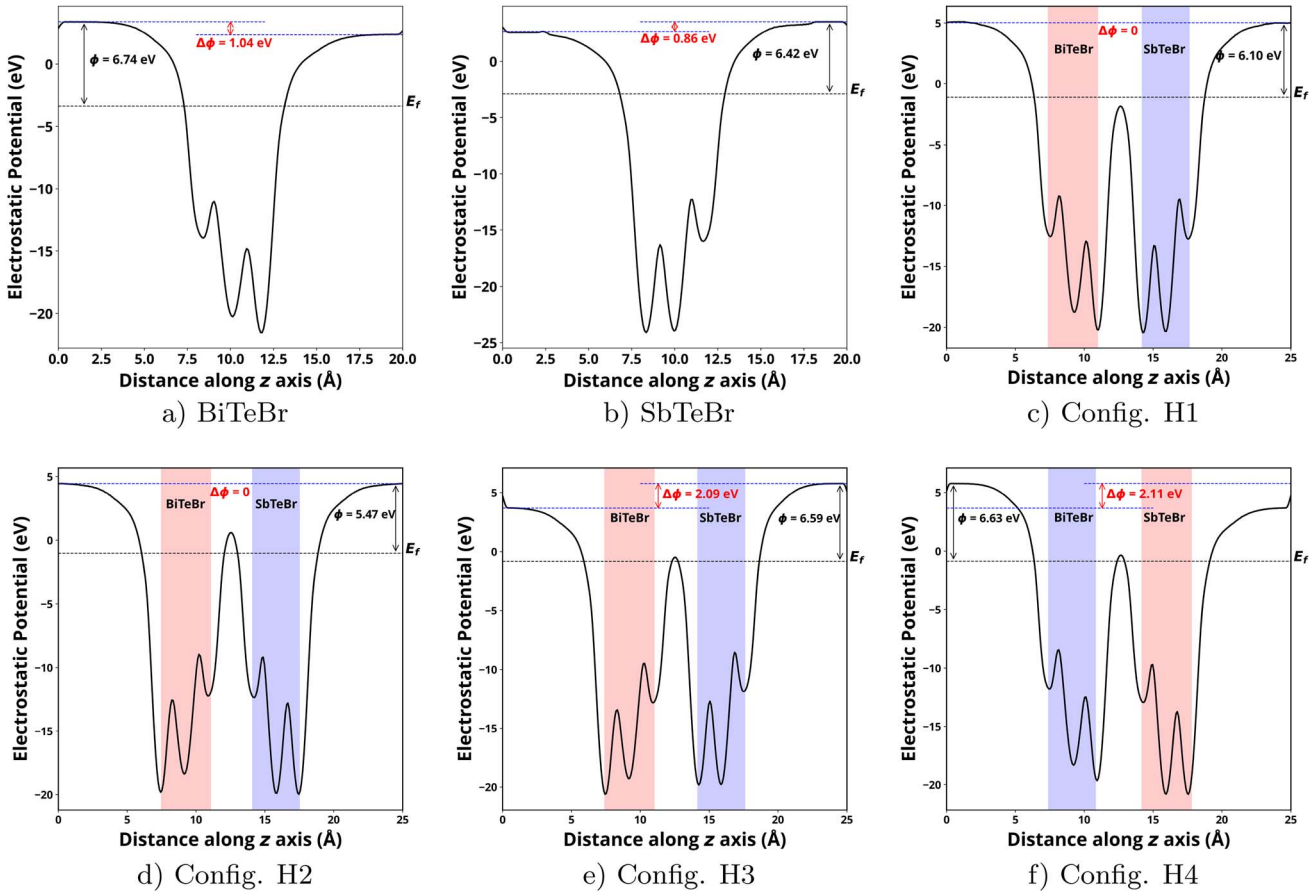

Fig. 8 Electrostatic potential difference with dipole corrections for monolayers: (a) BiTeBr and (b) SbTeBr along with heterostructures: (c) Config. H1, (d) Config. H2, (e) Config. H3, (f) Config. H4.

potential is defined as the slope of the variation in band edges with infinitesimal strain and can be expressed as $E_{i,j} = \frac{\Delta E_{i,j}}{(\Delta L_i / L_{i0})}$ where j represents the charge carrier type, *i.e.*, electron/holes (e/h) and i represents the direction either x or y, ΔL_i represents the change in the lattice parameter under the application of infinitesimal strain (±0.5% to ±2%) along the i direction, and $\Delta E_{i,j}$ denotes the change in the band edge energies in response to the strain applied by an infinitesimal amount. The mobility also depends on the elastic stiffness constant of the 2D monolayer, which we had previously calculated for the hexagonal cell using the SOECs given by eqn (3). However, for the orthorhombic cell, it is given by $C_{ii} = \frac{1}{A_0} \left. \frac{\partial^2 E_i}{\partial s_i^2} \right|_{s=0}$, where E_i is the change in total energy under the influence of infinitesimal strain in the i direction. Finally, on the basis of the deformation potential (DP) formalism, the longitudinal acoustic phonon-limited carrier mobility is expressed as

$$\mu_i = \frac{e\hbar^3 \left(\frac{5C_i + 3C_{i'}}{8} \right)}{k_b T m_d \sqrt{m_i} \left(\frac{9E_i^2 + 7E_i E_{i'} + 4E_{i'}^2}{20} \right)} \quad (5)$$

where $m_d = \sqrt{m_x m_y}$ and i are the transport direction either in x or y while i' is the direction perpendicular to it (y or x). The calculated mobility and effective mass along both directions, the DP constant, and the elastic stiffness constant are reported in Table 3. Both of these monolayers have very low effective mass, and because the mobility is related inversely to the square of the effective mass, we can see a high electron mobility in these monolayers. It is worth mentioning that higher mobility favors the transfer of charge carriers between the monolayers.

3.3 Band edge alignment and power conversion efficiency (PCE) in heterostructures

The band edge alignment in the heterostructure can be classified into three categories, namely the straddling gap (type-I), the staggered gap (type-II), and the broken gap (type-III), the schematics of which are shown in Fig. S8 of SI. The staggered gaps are necessary for photovoltaic applications because they can enhance the electron excitation and reduce recombination by accumulating electrons and holes in different monolayers, hence boosting the power conversion efficiency (PCE). In this band edge alignment, the VBM of the first material (M1) lies in between the band edge position of the second material (M2), while the CBM of M2 lies in between the band edge position of M1. As a result, an energetic offset is formed between two

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Table 3 Effective mass m_e and m_h; elastic constant C_{ii} (C_{11} and C_{22}); DP constant E_e and E_h corresponding to the conduction band (electrons) and valence band (holes); and mobility μ_e and μ_h along the x and y directions

	m_e (m_0)		m_h (m_0)		C_{ii} (N m^{-1})		E_e (eV)		E_h (eV)		μ_e (cm^2 V^{-1} s^{-1})		μ_h (cm^2 V^{-1} s^{-1})	
System	x	y	x	y	$i = 1(x)$	$i = 2(y)$	x	y	x	y	x	y	x	y
BiTeBr	0.205	0.193	−1.021	−0.852	28.74	28.71	−7.18	−7.14	−5.40	−5.60	292.29	311.25	21.43	25.22
SbTeBr	0.098	0.127	−1.155	−0.458	32.64	32.71	−8.94	−8.97	−4.98	−6.40	764.19	612.14	27.15	60.49
Config. 1	0.153	0.140	−0.567	−0.477	60.44	60.23	−8.82	−8.91	−6.82	−6.83	732.38	795.64	93.63	111.12
Config. 2	0.173	0.152	−0.625	−0.676	62.37	61.77	−9.45	−9.42	−7.26	−8.57	529.58	602.23	54.13	45.97
Config. 3	0.184	0.141	−0.914	−0.889	63.53	63.91	−8.36	−8.32	−7.28	−8.28	657.05	860.77	28.06	27.10
Config. 4	0.204	0.205	−0.788	−0.491	59.15	59.61	−9.00	−9.17	−7.82	−8.18	368.64	364.13	40.73	64.03

Table 4 Band gaps of monolayers and heterostructures determined using various functionals such as PBE, HSE06 and GW

	E_g (eV)		
System	PBE	HSE06	GW
BiTeBr	0.92	1.35	1.69
SbTeBr	1.11	1.62	2.13
Configuration H1	0.52	0.90	1.21
Configuration H2	0.86	1.29	1.23
Configuration H3	0.21	0.42	0.41
Configuration H4	0.56	0.81	1.33

material interfaces. At the time of solar irradiation, the photo-excitation takes place, while the electrons and holes are formed in both materials. The electrons that are excited for the material with a higher CBM (M1) will travel towards the material with a lower CBM (M2). Similarly, the holes in the material with a lower VBM energy (M2) are transferred to the material with a higher VBM energy (M1). This gathers electrons and holes in different materials, thus reducing the likelihood of recombination at the interface and enhancing the efficiency.

The percentage of solar energy that can be transformed and utilized is determined by PCE (η), which is based mainly on electron transfer between the monolayers. This metric is generally used to determine the efficiency of the heterostructure. Scharber[62] and Bernardi[63] have proposed an approach that is used to determine the upper limit of an excitonic solar cell. This is a widely used approach that has been employed to determine the photo-conversion efficiency (PCE) of 2D heterostructures[2] which was initially proposed for bulk materials,[64] and can be expressed as:

$$\eta = \frac{\beta_{FF} V_{OC} J_{SC}}{P_{solar}} = \frac{0.65\left(E_g^d - \Delta E_c - 0.3\right) \int_{E_g^d}^{\infty} \frac{P(\hbar\omega)}{\hbar\omega} d(\hbar\omega)}{\int_0^{\infty} P(\hbar\omega) d(\hbar\omega)}$$

where the maximum fill factor β_{FF} is taken to be 0.65, the maximum open-circuit voltage V_{OC} is determined as the difference between the band gap of the donor material E_g^d and the conduction band offset (CBO) ΔE_c subtracted by energy loss due to kinematics, which is approximated as 0.3 eV. The short-circuit current density (J_{SC}) is computed using an integral that takes into account the flux of the AM1.5 sun spectrum, assuming 100% external quantum efficiency. This integral takes into account photon energy, which spans from the donor's band gap to infinity. $P(\hbar\omega)$ is the spectral power density of the solar spectrum as a function of photon energy ($\hbar\omega$). P_{solar} is a normalization factor, which is calculated by integrating the entire AM1.5 solar spectrum ($P_{solar} \approx 1000$ W m^{-2}).

Two factors that influence PCE are open-circuit voltage and short-circuit current. The short-circuit current is determined solely by the donor band gap. The open-circuit voltage, which is mainly measured as the difference between the donor band gap and the conduction band offset, can also be stated as the difference in energy between the VBM of the donor material and the CBM of the acceptor material. The CBO must be less to have a higher V_{OC}, while the band gap of the donor material influences both the V_{OC} and J_{SC}. If E_g^d is higher, it will enhance V_{OC}, but it decreases J_{SC} by limiting the adsorption capacity of the entire solar spectra. Having E_g^d lower than 1 eV does not affect J_{SC} due to the absence of incident solar radiation in this range; therefore, for higher PCE, it is better to have values of E_g^d greater than 1 eV. In our work, all four combinations of heterostructure show type-II band alignment, which is a good candidate for photovoltaic devices. SbTeBr monolayer having a higher conduction band acts as the electron donor material in the first three cases, while in Config. H4, it acts as an electron acceptor due to a lower CBM value than BiTeBr. Thus, the arrangement of the monolayers is crucial to the electronic structure, which ultimately determines the PCE. The contour plot of PCE along with the band alignment is shown in Fig. 9. Although all four heterostructures are built with the same monolayer, Config. 1 shows that the direct band gap of the heterostructure has an overall band gap of 0.9 eV, with the donor band gap being 1.25 eV, while Config. H2 has a wider band gap as compared to the other configurations and also has a lower CBO, which is 0.27 eV giving rise to a higher PCE of 17.27%. Despite J_{SC} being slightly lower for this configuration due to the larger band gap of the donor material, V_{OC} compensated for this, which gives rise to a high PCE. Config. H3 overall has a very small band gap while having a high donor band gap. We can see a difference of 1.09 eV between the conduction band of SbTeBr and BiTeBr resulting in a lower V_{OC}, which in turn reduces the efficiency to a greater extent. Unlike the first three configurations, in which

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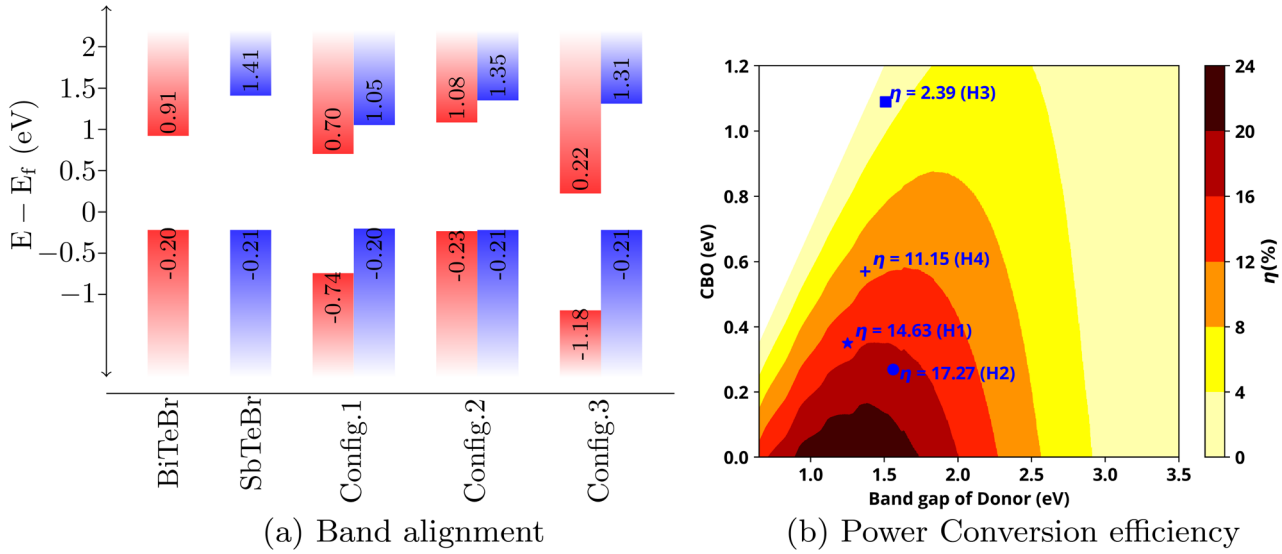

Fig. 9 (a) Band alignment of monolayers and heterostructures with Fermi level set to zero; the red colour in the heterostructure configurations represents BiTeBr while the blue colour represents SbTeBr. (b) Power conversion efficiency of heterostructures for different configurations.

SbTeBr acts as a donor material, in the last case BiTeBr acts as a donor. However, the efficiency is slightly low due to the high CBO, thus reducing V_{OC}.

3.4 MBPT based excitonic radiative lifetime determination for bright excitons

In this section, we focus on the optical response of monolayers and heterostructures by examining the influence of electron–hole (e–h) interaction. The Bethe–Salpeter Equation (BSE) is resolved in order to conduct this study. Our main goal is to investigate optical absorption spectra in the presence of in-plane light polarization. This investigation involves the analysis of first two bright direct excitons which possess greater oscillator strengths and thus dominate the optical response. The BSE optical absorption spectra, which are given by the imaginary part of the in-plane dielectric function, along with the transition of the first two excitons projected in the PBE band structure, are shown in Fig. 10 for monolayers and Fig. 11 for different heterostructure configurations. The analysis of the BSE spectra of BiTeBr shows the first two bright excitons at 1.18 eV and 1.23 eV respectively both being two-fold degenerate. The contribution for the exciton states corresponds to the transition between the two highest VB (*i.e.*, VBM and VBM-1) and lowest two CB (*i.e.*, CBM and CBM+1) at the Γ point for both the excitons, which can be visualized in Fig. 10(a1 and a2). Even in the case of SbTeBr, a monolayer, the transition for the first two bright excitons is found to involve both CB and VB, which are located at 1.74 eV and 1.97 eV respectively. However, the transition for the second bright exciton was slightly away from the Γ point, and the transition for the first bright excitons occurred at the Γ point.

The heterostructure with Config. H1 has a direct band gap and has a 3-fold degenerate bright exciton at 0.78 eV where the exciton transition takes place between two valence bands (VBM and VBM−1), which correspond to the SbTeBr monolayer, and two conduction bands (CBM and CBM+1), which correspond to the BiTeBr monolayer. The comparison with Fig. 7 allows assignment of these bands to specific monolayer components based on their characteristics. This arrangement belongs to the exciton called the interlayer exciton, where the electron–hole

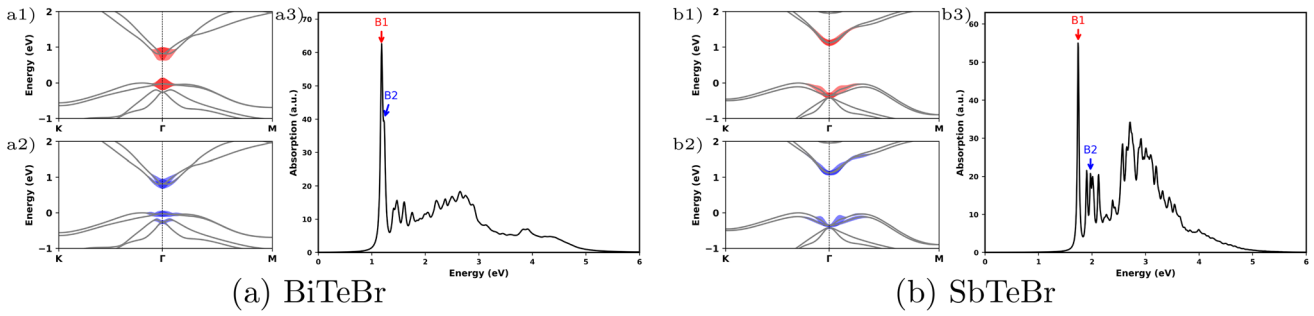

Fig. 10 Radiative processes that are associated with the first two bright excitons B1 and B2 (given in x1, x2 for B1 and B2, respectively). (x3) BSE absorption for monolayer (a) BiTeBr and (b) SbTeBr.

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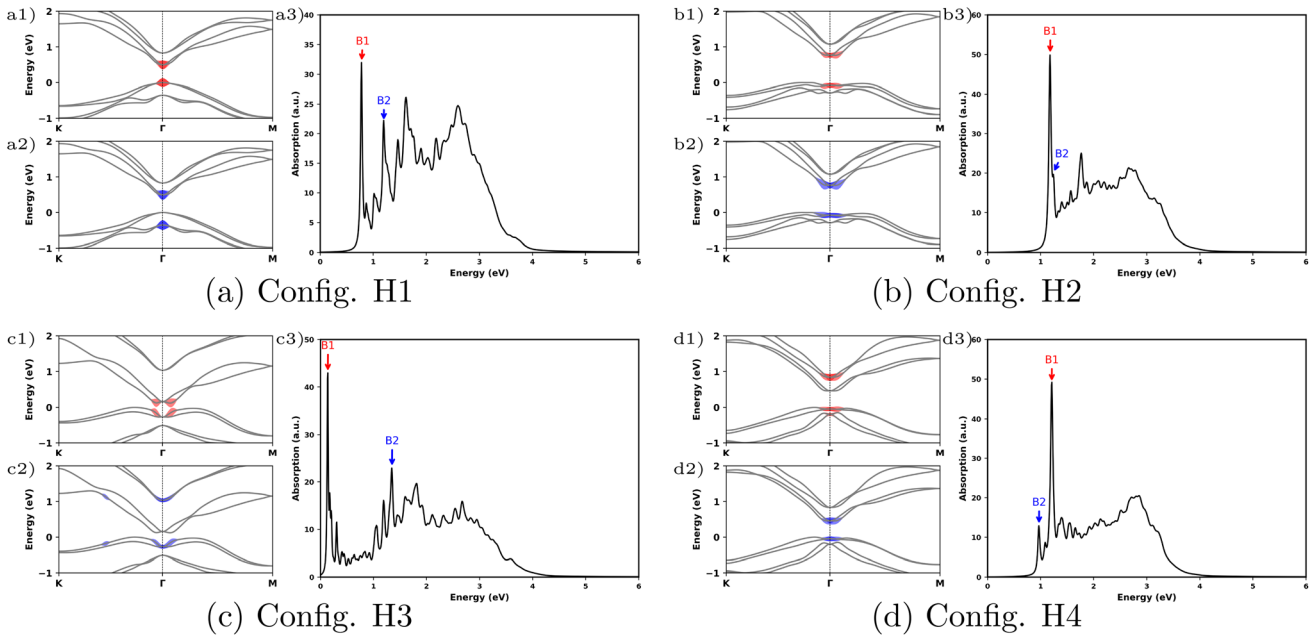

Fig. 11 Radiative processes that are associated with the first two bright excitons B1 and B2 (given in x1, x2 for B1 and B2, respectively). (x3) BSE absorption for heterostructures associated with different heterostructure (a) Config. H1, (b) Config. H2, (c) Config. H3 and (d) Config. H4.

pair is in two different layers of the heterostructure which constitute to form excitons. An interlayer exciton is a bound state between an electron and a hole in two distinct monolayers. They are linked to each other through their Coulomb attraction across the vdW gap. This spatial separation is what sets interlayer excitons apart from intralayer excitons, where both electrons and holes reside within the same atomic layer. For the second bright exciton, we observe an intralayer transition at 1.20 eV. Here, the transition involves the VB-2 and VB-3 valence bands and the two lowest conduction bands, all of which are primarily derived from the BiTeBr monolayer at the Γ point. Such intralayer excitons are characterized by both the electron and hole being localized in the same layer. The Config. H2 has the first two bright excitonic peaks located at 1.18 eV and 1.24 eV with the bright exciton being non-degenerate and the second bright being two-fold degenerate. The transition for the bright exciton takes place between two topmost VB and two lowest energy CB, while the transition for B2 is mainly between two topmost VB and one lowest CB near the Γ point. In Config. H3 the bright exciton B1 is two-fold degenerate located at 0.14 eV where the transition occurs at the highest VB which corresponds to SbTeBr and the lowest CB, which corresponds to BiTeBr near the Γ point giving rise to the interlayer transition, while the exciton B2 in Config. H3 has an interlayer transition occurring between VBM and two conduction bands that correspond to SbTeBr, *i.e.*, CBM+2 and CBM+3. Unlike the previous three heterostructures, the bright exciton B1 in Config. H4 is found at higher energy than the second bright exciton B2, both present at the Γ point. In this configuration, exciton B1 is found to undergo an intralayer transition between VBM and CBM, which belongs to the BiTeBr monolayer, while for exciton B2, the transition corresponds to the interlayer.

The exciton binding energy for the first two bright excitons is calculated as the difference in spectral peak position and direct gap between the corresponding GW band where the exciton transition occurs. The binding energies for monolayers and heterostructures are reported in Table 5. Note that the binding energy in heterostructures is found to decrease in comparison to monolayers. This helps dissociate electrons and holes easily, which enhances the efficiency of photovoltaic devices. Also, Config. H2 has the lowest binding energy among other heterostructures, which also has the highest PCE. Along with this, we have determined the exciton radiative lifetime $\tau_\lambda(0)$ of the exciton and λ which is the inverse of the decay rate $\gamma_\lambda(0)$ using the relationship given by:[9,65]

$$\tau_\lambda(0)^{-1} = \gamma_\lambda(0) = \frac{e^2 E_\lambda {\mu_\lambda}^2}{\varepsilon_0 \hbar^2 c a} \tag{6}$$

where e is the elementary charge, E_λ is the exciton energy, ${\mu_\lambda}^2$ is the square modulus of the BSE exciton transition dipole, ε_0 is the vacuum permittivity, c is the velocity of light, and a is the area of the unit cell. The average excitonic radiative lifetime at finite temperature T is given by:[65,66]

$$\langle \tau_\lambda(T) \rangle = \frac{3}{4}\left(\frac{2M_\lambda c^2}{{E_\lambda}^2}\right) k_B T \tau_\lambda(0) \tag{7}$$

where M_λ is the excitonic mass, which is obtained from the GW bands (Fig. S6 and S7 of SI) for light excitons, and k_B is the Boltzmann constant. We have analyzed the radiative lifetime of the first two bright excitons, which are most likely to have a shorter lifetime. The radiative lifetime, $\tau_\lambda(0)$, at 0 K, and the average radiative lifetime at room temperature (300 K) are tabulated in Table 5. The bright excitons in general have a shorter radiative lifetime since they undergo faster

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Table 5 Exciton energy, binding energy, and radiative lifetime at zero temperature ($\tau_\lambda(0)$) and room temperature ($\tau_\lambda(T)$) for the first two bright excitons B1 and B2 of monolayers and different heterostructure configurations

	BiTeBr		SbTeBr		Config. 1		Config. 2		Config. 3		Config. 4	
System	B1	B2	B1	B2	B1	B2	B1	B2	B1	B2	B1	B2
Exciton energy (eV)	1.18	1.23	1.74	1.97	0.78	1.20	1.18	1.24	0.14	1.35	1.21	0.97
Binding energy (eV)	0.57	0.51	0.82	0.58	0.43	0.54	0.21	0.14	0.45	0.31	0.50	0.46
$\tau_\lambda(0)$ (ps)	0.20	0.34	0.13	0.35	0.44	0.54	0.15	0.51	1.71	0.67	0.32	1.38
$\tau_\lambda(T)$ (ns)	0.75	1.18	0.23	0.50	7.82	4.40	0.42	18.15	76.92	7.68	2.95	84.42

recombination, whereas other excitons will have a longer lifetime than the first bright exciton. The $\tau_\lambda(0)$ values are in the ps range, although the radiative lifetime at room temperature is in the ns range. BiTeBr was found to have a longer radiative lifetime compared to SbTeBr. In general, heterostructures have a longer radiative lifetime when compared to monolayers, which is favorable for photovoltaic applications. In spite of having higher PCE and lower binding energy, this is favorable for photovoltaic applications. The Config. H1 and H3, where the exciton transition is interlayer, have a longer radiative lifetime for bright excitons. This is because the electron–hole pair is spatially separated across different monolayers, which reduces the oscillator strength of radiative recombination and consequently enhances the exciton lifetime.[67–69] Even though Config. 2 forms type-II alignment, the direct band gap, located at Γ corresponding to the BiTeBr monolayer where the exciton transition takes place, has a shorter radiative lifetime compared to other heterostructure configurations. This is because the direct gap corresponds to the BiTeBr monolayer; this favors the intralayer exciton being formed, which also has a similar lifetime of monolayers. The Config. H4 also has intralayer excitons formed, resulting in shorter lifetimes compared to Config. 1 and Config. 3 heterostructures, which have interlayer excitons.

4 Conclusions

The 2D Janus monolayer of BiTeBr and SbTeBr, and the corresponding construction of four different heterostructures, are investigated systematically using first-principles DFT based calculations coupled with many-body perturbation theory formalism. These materials have been discovered to be stable, with thermodynamic, mechanical, and dynamic stability showing favorable conditions. These monolayers also show a minimal lattice mismatch, which favors the formation of heterostructures. The electronic properties are considered, such as the appropriate band gap to absorb the solar radiation and high charge carrier mobility, which favor the transport of electrons and holes, thus making it suitable for the photovoltaic application. Depending upon the interface atoms that participate in the vdW interaction between the monolayers, four different combinations of heterostructures are formed, which are found to be stable. Due to the difference in charge distribution in these heterostructures, it affects their electronic and optical properties. Even though all heterostructures form type-II band alignment, their PCE varies by a great extent in the range between 2.39% and 17.27% due to the variation in band edge position. The binding energy of the heterostructure is noted to decrease, which is beneficial for photovoltaic devices. The interlayer bright excitons that are formed in the heterostructure have a longer lifetime, which can reduce radiative decay, and the electrons and holes can be separated and transported. This work emphasizes the effect of interfacial atomic arrangement, which could govern not only the possible Rashba spin texture, but also the excited state properties like excitonic radiative lifetime and type-II band edge alignment favorable for solar cell applications. The sensitivity of the ground state and excited state properties of the heterostructure to the terminating atoms at the interface would certainly provide a roadmap of fine tuning such properties under experimental synthesis conditions as well.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the SI. See DOI: https://doi.org/10.1039/d5ta04076k.

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